

DBMS PRACTICAL REPORT

ENTITY-RELATIONSHIP DATABASE DESIGN FOR A STUDENT MANAGEMENT SYSTEM

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Registration Number

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1. Introduction

This project involves the design and implementation of an Entity-Relationship database for a Student Management System using MySQL. The database is designed to manage student records, courses, lecturers, enrollments, and assignments.

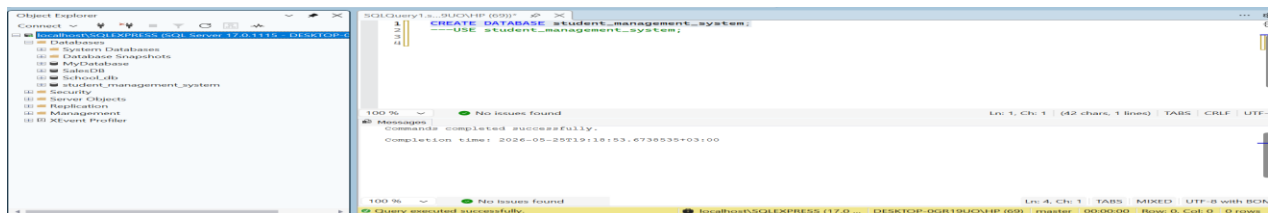
The project demonstrates database creation, table relationships, data insertion, SQL queries, and advanced database features.

2. Database Created

Database Name

student_management_system

Screenshot:



3. Tables Created

The following tables were created:

- Student
- Lecturer
- Course
- Enrollment
- Assignment

4. Relationships Used

The following relationships were implemented:

- One lecturer can teach many courses
- One student can enroll in many courses
- One course can have many assignments

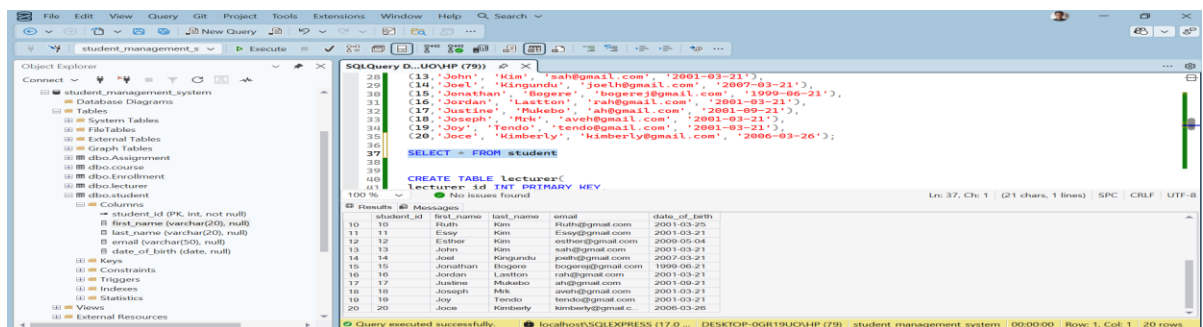
Foreign keys were used to establish relationships between tables.

5. Inserted Records

Sample records were inserted into all tables.

These include:

- Student information
- Lecturer information
- Course details
- Enrollment records
- Assignment records



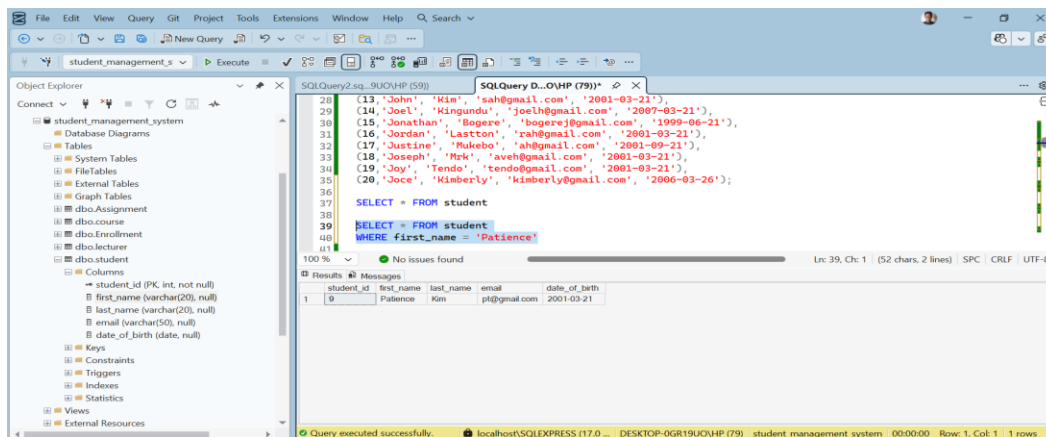
6. SQL Queries

SELECT Query

SELECT * FROM student;

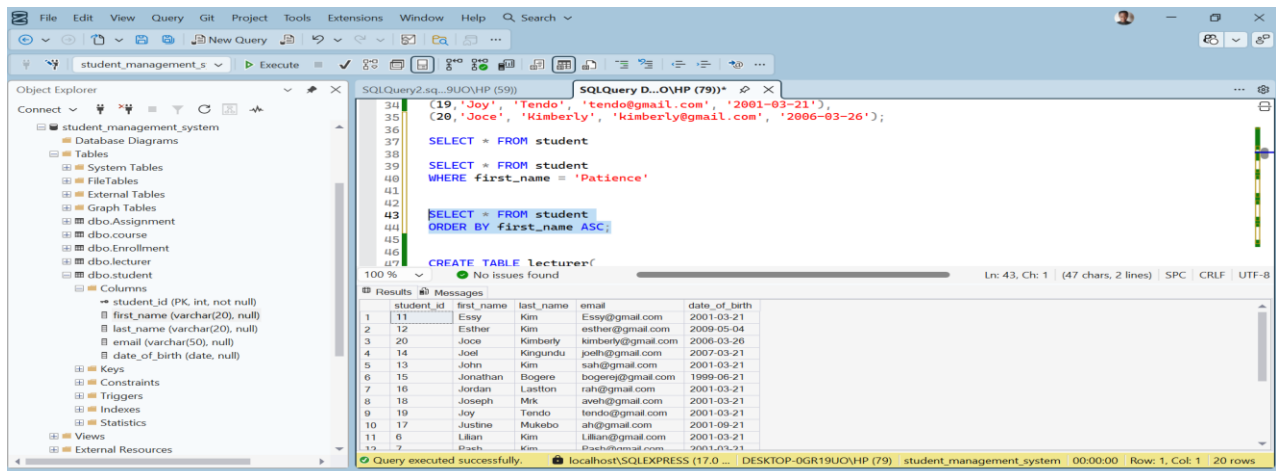
WHERE Clause

SELECT * FROM student
WHERE first_name = 'Patience';



ORDER BY

SELECT * FROM student
ORDER BY first_name ASC;



Aggregate Functions

COUNT()

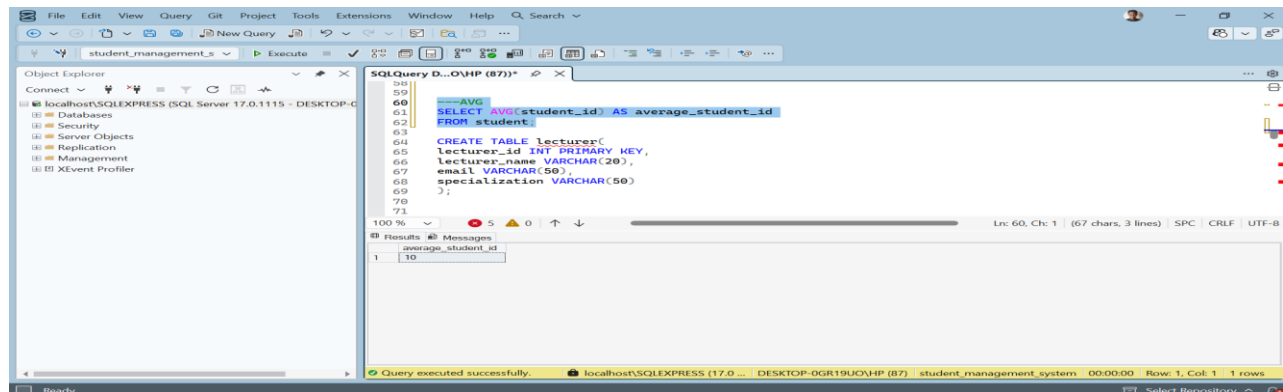
```
SELECT COUNT(*) FROM student;
```

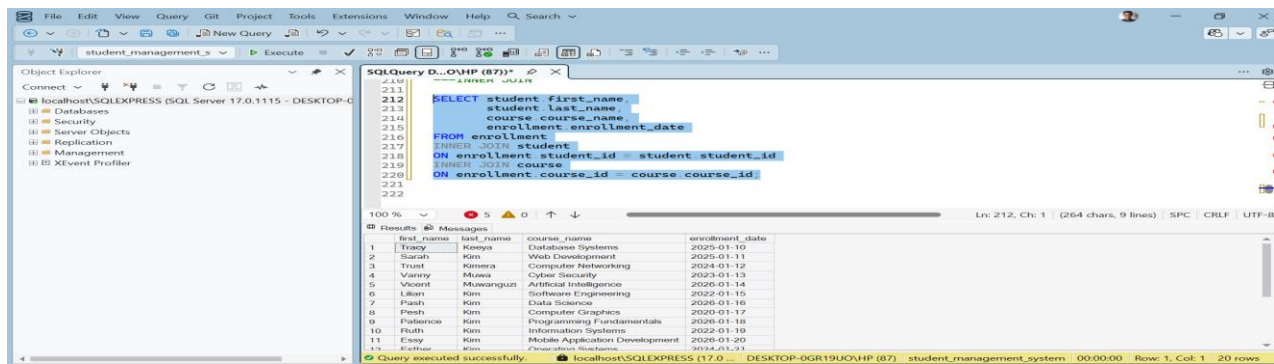
SUM()

```
SELECT SUM(student_id) FROM student;
```

AVG()

```
SELECT AVG(student_id) FROM student;
```





6. UPDATE Statement

```

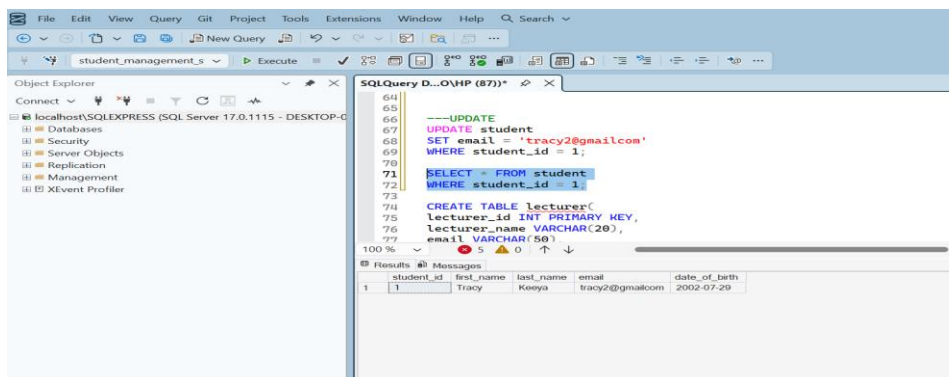
UPDATE student
SET email = 'tracy2@gmailcom'
WHERE student_id = 1;

```

```

SELECT * FROM student
WHERE student_id = 1;

```



7. Advanced Database Feature

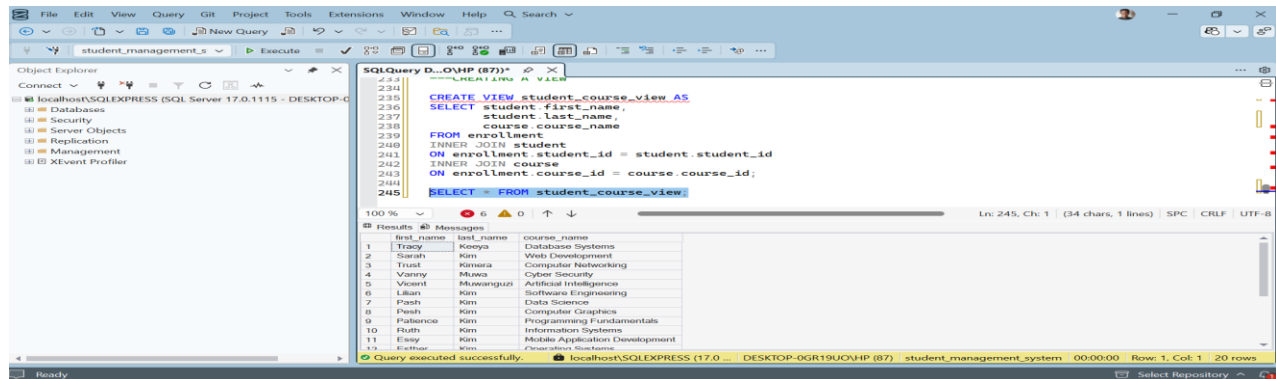
A VIEW was created to display student names together with course names.

```

CREATE VIEW student_course_view AS
SELECT student.first_name,
       student.last_name,
       course.course_name
FROM enrollment
INNER JOIN student
ON enrollment.student_id = student.student_id

```

INNER JOIN course
ON enrollment.course_id = course.course_id;



8.Conclusion

The project successfully demonstrated the design and implementation of a Student Management System Database using MySQL. Tables, relationships, queries, and advanced database features were implemented successfully.